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The ecology and evolution of microbial metabolic strategies

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Free-living microbes are generally capable of growing on multiple different nutrients. Some of those nutrients are used simultaneously, while others are used sequentially. The pattern of nutrient preferences and co-utilization defines the metabolic strategy of a microorganism. Metabolic strategies can substantially affect ecological interactions between species, but their evolution and distribution across the tree of life remain poorly characterized. We discuss how the confluence of better computational models of genotype-phenotype maps and high-throughput experimental tools can help us fill gaps in our knowledge and incorporate metabolic strategies into quantitative predictive models of microbial consortia.

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Introduction

The complexity and size of microbial metabolic networks creates latent metabolic capabilities [1], and makes it easy to acquire novel metabolic traits, for instance by facilitating the attachment of horizontally transferred genes to the existing network. As a result, most free-living bacteria are capable of growing on multiple environments and using many different types of nutrients [1]: they are metabolic generalists. Due in part to this widespread generalism, species within an ecosystem often have a substantial overlap in the nutrients they may use for growth [2]. In principle, this could lead to resource competition. However, not all of these generalists compete for all of the available resources at the same time [3]. Rather, microbes often use nutrients sequentially, activating their metabolic pathways one at a time according to a hierarchy of preferences coded in their gene regulatory networks [4].

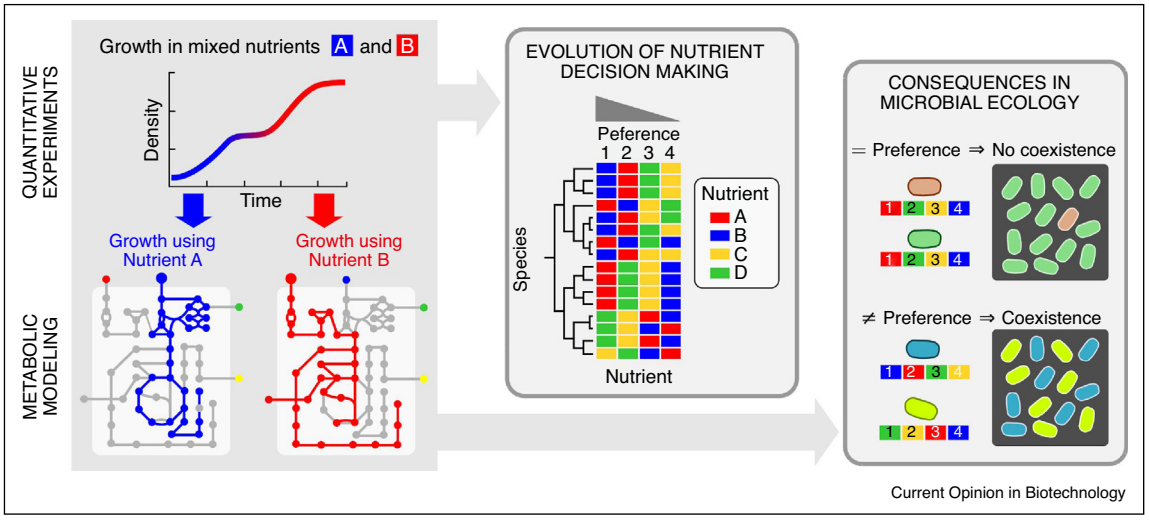
The existence of nutritional 'preferences' (the order in which multiple substitutable resources are consumed) has been known since the seminal work of Jacques Monod on diauxic shifts in the 1940s and 50s. Monod's discovery that the bacterium *Escherichia coli* prefers to use glucose over lactose when both are available played a foundational role in molecular biology. It launched the entire field of gene regulation, which started off by characterizing in detail the molecular mechanisms and genetic circuitry (the lac operon) that allows *E. coli* to implement this particular metabolic strategy. More recently, the lactose switch has played a similarly important role in systems biology, providing a blueprint for the development of quantitative biophysical models of gene regulation [5].

Despite the pivotal place that metabolic decision-making has played in modern biology and the ample body of work on model organisms, many basic fundamental questions remain about the evolutionary origin and diversity of carbon source utilization strategies across the bacterial tree of life [3,6]: How evolutionarily conserved are carbon source utilization strategies? How evolvable are they, and how are they coded in the genotype-phenotype map? Can we predict them from genomic data and first metabolic principles? What fraction of the nutrients are typically under regulatory control? And how do nutritional preferences affect community assembly and coexistence? These questions have become a subject of growing interest, and our aim in this piece is to review recent relevant work and stress the most important gaps in our knowledge in this area (Figure 1).

Hierarchical nutrient preferences and co-utilization in microbial populations

The preference of *E. coli* for glucose over lactose when both are available at high concentrations extends to most other sugars and sugar alcohols. A hierarchical preference has been reported also among those 'secondary' sugars (those that *E. coli* chooses to ignore when glucose is available), with lactose at the top followed by arabinose, xylose, sorbitol, rhamnose and ribose, in that order [7]. In addition to sugars, *E. coli* also prefers glucose over non-sugars such as acetate. A particularly insightful example of sequential nutrient utilization is found in overflow metabolism: when grown under aerobic conditions in the presence of high concentrations of glucose as the only carbon source, *E. coli* uses metabolically cheaper but inefficient fermentative

Figure 1



A combination of quantitative high-throughput experiments and predictive metabolic models can help us map the genotype-phenotype map of microbial metabolic strategies. A main goal of systems biology is to predict these strategies from genomic information. This will allow us to understand the evolvability of metabolic decisions, elucidate how these decisions differ over evolutionary timescales, and understand their implications for microbial coexistence in ecological communities.

pathways to metabolize glucose, resulting in the production and secretion of excess acetate that will be used only after the glucose has been depleted, following a diauxic shift [8].

The existence of carbon source preferences can be explained by the same resource allocation problem that is behind overflow metabolism: having only a limited amount of internal resources (e.g. proteome), organisms will allocate them to the pathways and nutrients that minimize the overall cost/benefit ratio. Metabolic models that incorporate global resource allocation constraints, most of them based on Flux Balance Analysis, successfully predict overflow metabolism [8–12], and some of them also nutrient use preferences [11]. While they are all based on the same principle of imposing a total flux budget, these models consider different underlying mechanisms, such as molecular crowding, membrane crowding or limited total proteome size. It remains an open question to what extent, and under which conditions, each of these mechanisms would be prevalent [13•].

Altogether, these models highlight that sequential utilization of carbon sources may be a response to the presence of metabolic tradeoffs. Sequential utilization appears to be common, at least during transient nutrient pulses (less so under constant environments, for example, in chemostats [14]). However, even under those circumstances not all carbon sources are necessarily consumed sequentially, and many are co-utilized [3]. Predicting which nutrients will be co-utilized and which will be sequentially used is an area of growing interest, and recent studies suggest that metabolic network topology could be key [15••,16]. Encouragingly, recent FBA-based models have been able to successfully predict the metabolic strategies of *E. coli* under a range of different mixed carbon-source environments [15••,17••,18].

One challenge for these models is that the choice between sequential use and co-utilization is governed by resource concentration, with the latter being common when nutrients are present at low abundance and the former being most prevalent at high nutrient concentration (see for instance [3] and references therein). Moreover, in conditions where nutrients are used sequentially, the rank of metabolic preferences is also contingent on resource concentration and may be reversed by modulating nutrient availability [4,19•]. The effect of nutrient concentration on metabolic strategies has been recently addressed by several quantitative computational models [10,15••,16]. Although the focus has been on *E. coli*, the computational tools developed in those studies are based on Flux Balance Analysis [20], a flexible modeling framework that can be applied to any metabolic network. Therefore, these studies could be, in principle, extended to other microbes and nutrient environments in a straightforward manner.

Computational models have also primarily focused on metabolic decisions between pairs of carbon sources. However, these simple nutrient mixes seldom represent the complex environments bacteria typically find themselves into [3,21]. Is it possible to predict, from first principles, how carbon source utilization preferences change when more than two nutrients are present? At

high concentrations, complex utilization patterns are possible in order to optimize growth. In principle, a given microorganism may grow better on a preferred nutrient than on either one of two alternative 'secondary' nutrients. Yet, when both of those 'secondary' nutrients are available, co-utilizing them may synergically produce stronger growth than that on the 'primary' resource alone. Bacteria are equipped to perform complex computations with more than two inputs [22], so they could alter their metabolic choices in complex nutrient environments relative to what we may expect based on pairwise carbon source mixtures. More experimental work is needed to elucidate to what extent pairwise carbon source decisions and hierarchies are predictive of those in more complex environments.

A current limitation of the predictive models discussed above is that they operate at the population rather than at the single-cell level. At intermediate or low nutrient concentrations the preferred carbon source at the single-cell level may depend on the cell's previous metabolic state [23–25]. Thus, a population may co-utilize nutrients even though none of the single cells in the population follow that strategy [14]. There may be many plausible evolutionary explanations for single-cell heterogeneity in metabolic strategies, for example, this could represent a bet-hedging strategy [25]. Intriguingly, external nutrient concentration can modulate the proportion of cells using different substrates from a mixture. This suggests that simultaneous and sequential utilization might be only two of the strategies lying on a trade-off between taking advantage of the current and preparing for the next-best substrate available, once the 'best' one is depleted [26,27]. Modeling strategies that predict metabolic heterogeneity in populations at the single-cell level are an area of growing interest [14,25,28,29].

Besides *E. coli*, carbon source preferences have also been studied in other species, mainly for clades containing model organisms such as Firmicutes (e.g. *Bacillus subtilis*) or Pseudomonadaceae [30], as well as for certain soil or gut bacteria of interest (e.g. Ref. [31]). The hierarchy rank for members of the *Pseudomonas* genus appears to be different from *E. coli*'s, and glucose is not a top choice: organic acids such as succinate or citrate [3], aromatic compounds, and even various amino-acids are preferred over glucose in mixed carbon media [30]. However, the data on carbon source preferences for non-model organisms remains fragmented and scarce, and our knowledge of carbon source preferences and utilization strategies across the bacterial tree of life is still very limited.

Evolution and evolvability of microbial metabolic strategies

There is an emerging interest in the phylogenetic conservatism of phenotypic traits in microbes [6]. The decision between co-utilizing two carbon sources or using them sequentially is governed by the dynamic interplay between metabolic and gene-regulatory networks [15••,16,32]. This suggests that both networks must co-evolve in order for bacteria to implement 'optimal' metabolic strategies in complex nutrient compositions. For example, the evolution of the metabolic network by the addition of a new reaction must be followed by the evolution of the adequate regulation of that reaction in order to optimize its physiological output. Despite their importance, very little is still known about the evolution and evolvability of microbial metabolic strategies [33,34]. Clearly, the topology of metabolic and genetic networks should play a critical role. Their robustness to evolutionary changes, for instance through the acquisition of new transcriptional regulators or metabolic genes by Horizontal Gene Transfer [1], should dictate how stable metabolic strategies are over evolutionary timescales, and how different we should expect them to be among closely related organisms.

Investigating this question requires an integration of metabolic and genetic networks into a single quantitative model of the genotype-phenotype map [35]. Recent computational work connecting carbon source utilization strategies to the topology of metabolic and macromolecular expression networks [15••,17••] provides a promising starting point. The confluence of advances in dynamic FBA [36,37] with the ready availability of genomic data and of annotation tools that can be used to reconstruct gene regulatory-networks [22], makes us optimistic that the answers to most of the questions posed above are within reach.

Beyond computational models, experimental approaches will also be necessary to understand the evolution of nutrient preferences in bacteria. Plate readers have become inexpensive and have the capability to perform hundreds or even thousands of diauxic assays in parallel. They could be thus deployed to investigate the nutrient preferences of large collections of organisms. Experimental evolution can also offer insights into the evolvability of nutrient preferences. Recent work in *E. coli* suggests that the hierarchy of metabolic decisions is evolutionarily robust, at least over short timescales. Preference hierarchies among pentose sugars remained stable after ~2000 generations of adaptation to a wide range of mixed carbon source environments [38]. Extending such a study to a larger number of species and conditions would require the automation of experimental evolution and a significant increase in its throughput. Novel robotic liquid handling and miniature chemostats are being developed that may make this possible in the near future [39•].

Ecology of microbial metabolic strategies

Many habitats on earth harbor an enormous microbial diversity. This finding has motivated a growing interest in understanding the ecological mechanisms that mediate microbial coexistence [40–44]. One of the classic

mechanisms that supports coexistence is specialization, which allows species to split and share the niche space [45,46]. For example, multiple consumers may coexist in a mixture of substrates provided that they are limited by different ones [47]. Carbon source utilization preferences can potentially play a very important role by temporally partitioning resources among species: Even two strains that had perfect overlap in the types of carbon sources they can use for growth, may still avoid competition if their hierarchical preferences for those carbon sources were different [46]. In other words, hierarchical carbon source utilization turns metabolic generalists into serial metabolic specialists, separating their niches in time. Theory has shown that this may allow them to coexist even in simple unstructured environments. First, models that incorporated adaptive (i.e. sequential) metabolic strategies into a classical MacArthur consumer-resource model [48] have been found to explain coexistence of metabolic generalists in serial batch culture environments such as those described above [40]. The coexistence of large numbers of metabolic specialists has also been found within the framework of the 'Stable Marriage Problem' [49•].

Predicting microbial coexistence in a given environment is an area of growing interest in systems biology and biotechnology [50]. A better understanding of how metabolic strategies are coded in the genotype of a species is critical for that purpose. Recent success in creating models of metabolic networks from genomic data [51] offer hope that accomplishing this goal may be within reach in the coming years. As before, a close integration of quantitative experiments and theory will be key [37,40].

Conclusion

We have argued in this short piece that in order to develop quantitative and predictive models of microbial communities, and to understand the patterns of biodiversity found in most natural habitats, it is crucial to understand the evolution, distribution and diversity of bacterial nutritional preferences and metabolic strategies across the tree of life [52]. This is an area where microbial ecology, evolution and behavior may interact in complex ways, as they all can occur at similar timescales [53]. For instance, if metabolic decisions were easily evolvable this may have consequences for the strain-level structure of microbial populations [54], as it would determine whether two closely related bacteria may avoid competitive exclusion [48,49•]. Besides its conceptual and scientific importance, addressing this matter is also of clear practical importance, for instance if we wish to develop evolutionarily stable and ecologically resilient microbial consortia for biotechnology applications.

Finally, we suggest that introducing nutrient preferences into models of community assembly highlights the need to define a metabolic niche in a quantitative, rather than a qualitative manner (e.g. the list of nutrients a microbe may use, which is often use as a proxy for the metabolic niche of a species, could fail to predict whether two microbes will coexist or not). Recently introduced algorithms have made it possible to predict the 'seed set' of extracellular metabolites that a metabolic network may use to grow [2]. Combining these approaches with computational methods that allow us to infer nutrient preferences from metabolic networks [15••,17••,18], and with algorithms that automate the inference of metabolic networks from genomes [51], we believe that we are close to producing a computational pipeline to predict microbial coexistence from genomic data. This may also teach us how to design synthetic environments where two or more species will coexist [55], thus providing a stepping stone towards the ultimate goal of engineering robust and efficient microbial consortia.

Conflict of interest statement

Nothing declared.

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